

IssueOps

CONTINUOUS AI FOR GITHUB ISSUES



PROJECT PRESENTATION

NOVEMBER 2025

VAAS

DS252

This course provides a hands-on introduction to cloud computing, covering core distributed systems concepts, cloud architectures, service models, storage, serverless design, DevOps, and advanced topics such as MLOps, edge-cloud systems, and AI-driven workflows. Students learn through a blend of instructor guidance, AI-based teaching agents, peer collaboration, and weekly labs. The course culminates in a team capstone project where AI agents assist in designing and deploying a full cloud application.

VINAY

<https://github.com/vinaaaaay>

ASHWIN KM

<https://github.com/AshwinKM103>

SAI HARSH

<https://github.com/SaiHarsh24>

Note: One team member withdrew from the course midway through the project, which resulted in an additional “A” appearing in the team name.

PROBLEM

1. HIGH VOLUME AND NOISE

Large repositories receive hundreds of issues; many are duplicates, unclear, or irrelevant, making triage slow and error-prone.

2. MANUAL LABELING

Assigning labels, estimating severity, and routing issues to the right maintainers consumes substantial time.

3. MAINTAINER BURNOUT

The continuous stream of issues overwhelms developers, reducing focus on actual feature development and bug fixing.

7.3 HOURS LOST PER WEEK PER DEVELOPER TO TRIAGE, DUPLICATES, AND NOTIFICATIONS

3+ DAY DELAYS BEFORE ISSUES ARE EVEN CATEGORIZED OR ROUTED

59% OF MAINTAINERS CONSIDER QUITTING DUE TO BURNOUT AND OVERHEAD

“ We already burn countless unpaid hours triaging issues and dealing with duplicates instead of writing code ”

-Mike McQuaid, Lead Maintainer of Homebrew

SOLUTIONS AVAILABLE

Problem	Github Copilot AI Triage	Dosu	Linear
Spam Filtering	✗	✓	✗
Auto Issue Labels	✗	✓	⚠
Severity Estimation	✓	⚠	✗
Component Routing	✗	⚠	✓
Expertise Based Assignment	✗	✗	⚠
Availability Awareness	✗	✗	✗
Workload Fairness	✗	✗	⚠

OUR SOLUTION

WHAT IT IS

A GITHUB APP & STREAMLIT DASHBOARD SOLUTION



DS252-IssueOps

GitHub App

[Configure](#)

Manage your installation settings.

An app that automatically sees issues, filters out duplicates and spam, classifies the issues and assigns them to a suitable developer.

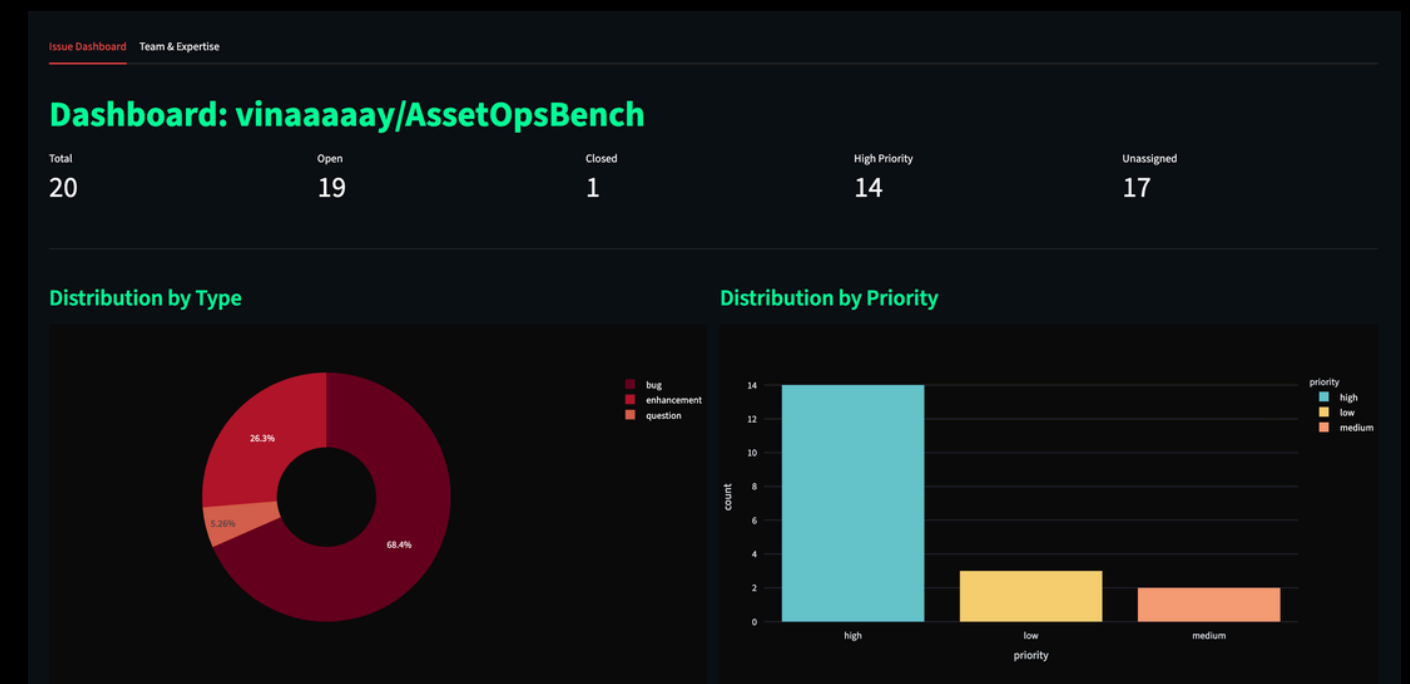
Developer

 SaiHarsh24

[Website](#)

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OUR SOLUTION

EXPLAINED

1. HIGH VOLUME AND NOISE

Llama 3-based spam filtering instantly blocks low-quality issues, while LLMs rewrite vague titles to ensure clarity.

2. MANUAL LABELING

Mistral 7B analyzes context to automatically assign Priority, Component (e.g., Frontend/Backend), and Type (e.g., Bug/Question/Enhancement) in real-time, removing manual overhead.

3. MAINTAINER BURNOUT

The "Master Agent" routes tasks based on actual code expertise and contributor working hours (availability), ensuring issues are assigned only to the right person at the right time.

OUR SOLUTION

GETTING STARTED WITH

1. INSTALL THE GITHUB APP

Go to the installation page, install the app, and select the repositories to manage. The system will automatically scan for new issues every hour.

2. OPEN THE DASHBOARD

Access the dashboard to view analytics, monitor issue distributions, and manage team expertise.

3. RUN MANUAL SCANS WHEN NEEDED

Use the dashboard's manual trigger to initiate immediate issue processing whenever required.

OUR SOLUTION

IN ACTION

1.

☐ Open 40 Closed 3

☐ ☒ This function is crashing the memory spam
#41 · by vinaaaaay was closed 3 days ago

☐ ☒ The function is crashing the memory spam
#16 · by vinaaaaay was closed 3 weeks ago

☐ ☒ buy solana spam
#11 · by SaiHarsh24 was closed 3 weeks ago

2.

☐ Open 40 Closed 3 Author Labels Proj

☐ ☒ multivariate anomaly detection backend bug High SLA Breached
#43 · vinaaaaay opened 3 days ago

☐ ☒ Center Align Signin Form on Web Portal enhancement frontend low
#42 · vinaaaaay opened 3 days ago

☐ ☒ ui not working bug frontend High SLA Breached
#40 · SaiHarsh24 opened last week

☐ ☒ UI Bug bug frontend High SLA Breached
#39 · SaiHarsh24 opened last week

☐ ☒ driver trip direction is not updating correctly at start backend bug High SLA Breached
#38 · SaiHarsh24 opened last week

☐ ☒ Driver Trip Direction Not Updating Correctly on Start backend bug High SLA Breached
#37 · vinaaaaay opened last week

3.

Re: [vinaaaaay/AssetOpsBench] driver trip direction is not updating correctly at start (Issue #38) Inbox x

**ds252-issueops[bot]** <notifications@github.com> [Unsubscribe](#)

to vinaaaaay/AssetOpsBench, me, Assign ▾

Assigned [#38](#) to [@vinaaaaay](#).

—

Reply to this email directly, [view it on GitHub](#), or [unsubscribe](#).
You are receiving this because you were assigned.

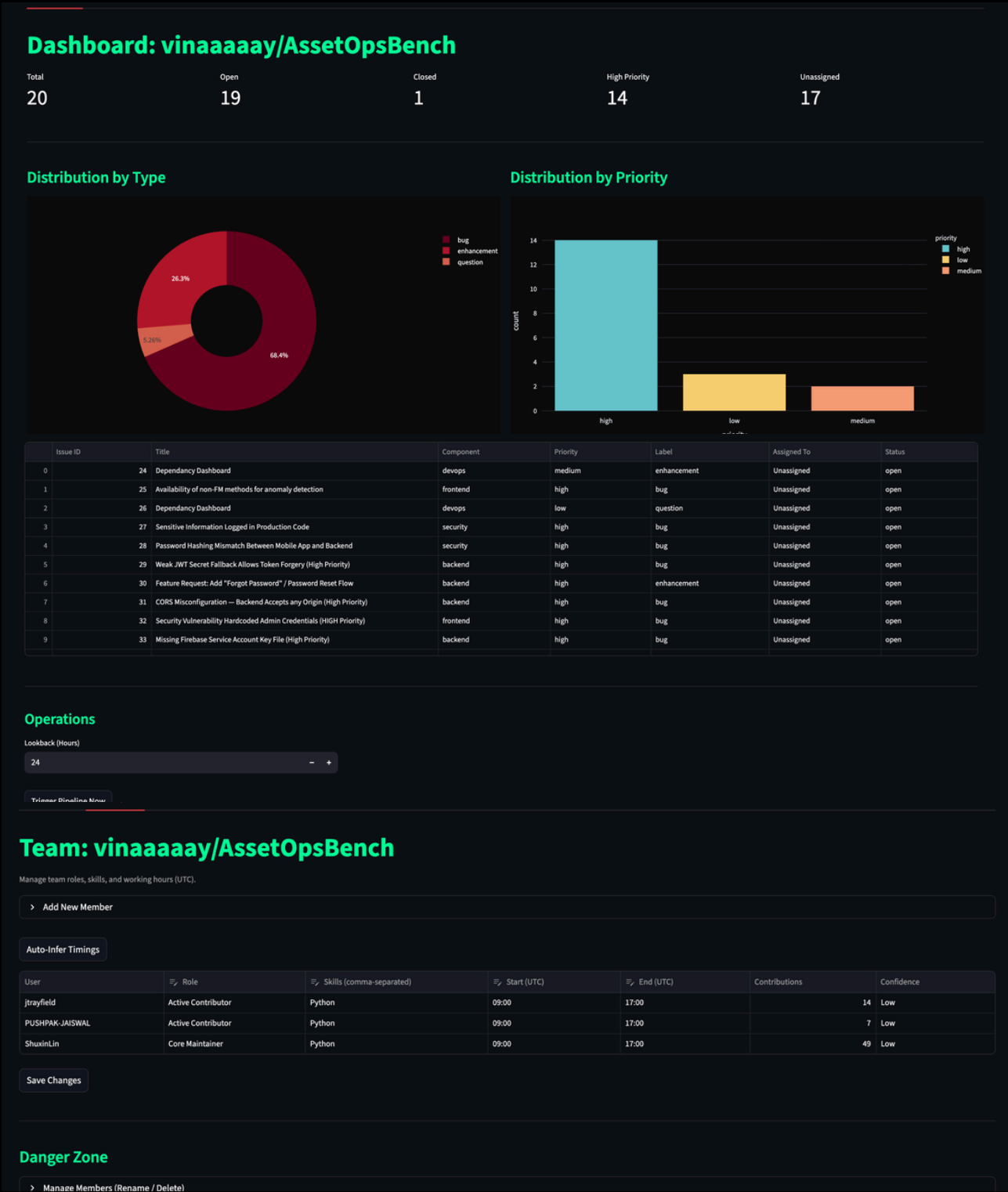
Wed, Nov 19, 7:00 PM (10 days ago) ☆ 😊 ↩ ⋮

DASHBOARD

Presents an at-a-glance summary of repository health.

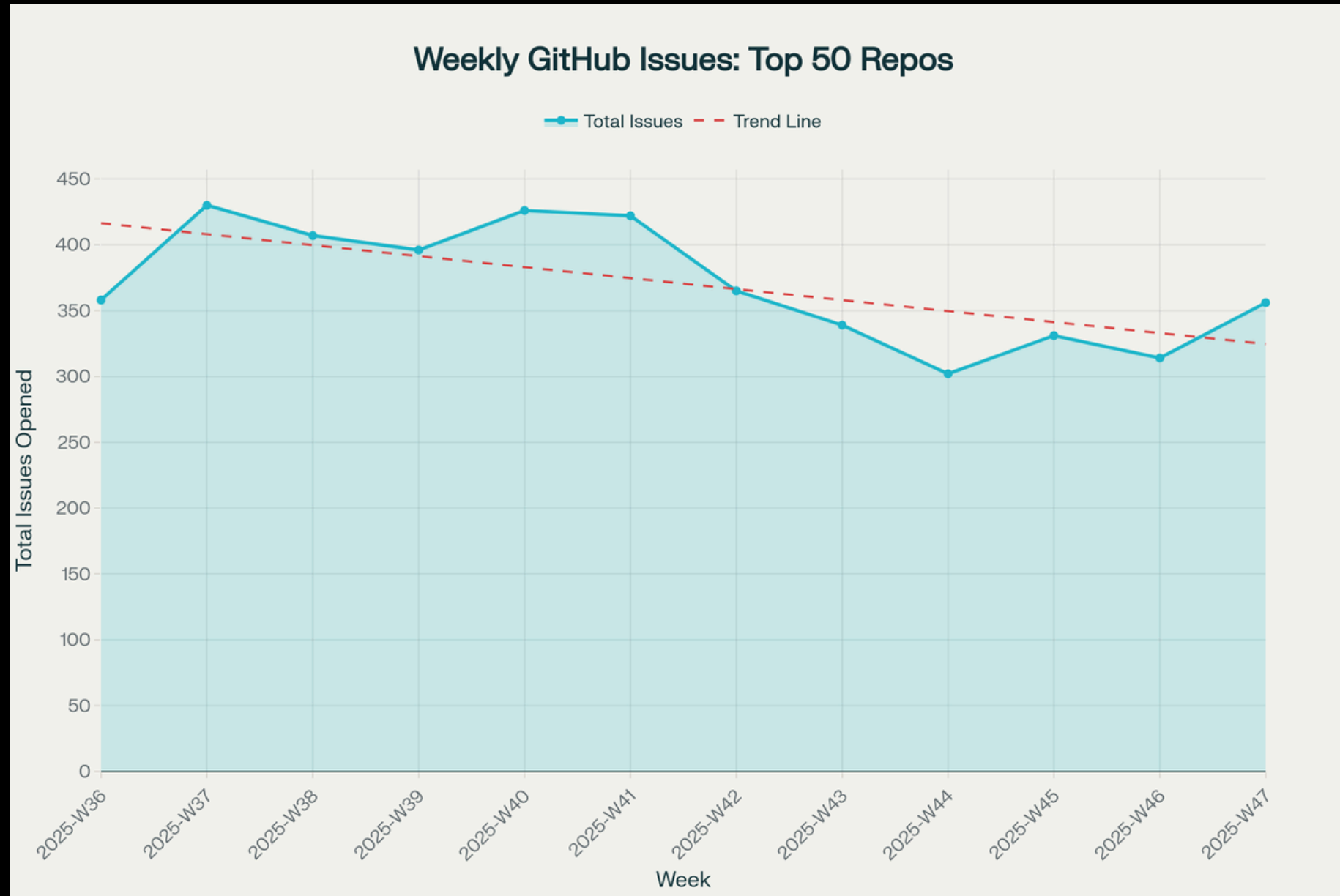
Displays issue-level metadata.

Shows the inferred team structure, contributor roles, skill sets, and working hours for the repository.



OUR SOLUTION

METRICS



COST:

- 2.5-3\$ PER THOUSAND ISSUE
- ~10\$ A MONTH FOR THE EC2 INSTANCE

TIME:

- 2-5 MIN FOR 10 ISSUES PER REPO
- BATCH PROCESSING BUT HAS COLDSTARTS
- HAPPENS IN PARALLEL FOR MULTIPLE REPOS

ACCURACY:

- INITIALLY WE USED A TUNED BERT MODEL ON THE AIC PAPER DATASET
- IT PROVIDED >85% ACCURACY ON THE TEST DATA
- BUT IT WAS NOT UPTO MARK DURING PEER TESTING, SO WE SWITCHED TO MISTRAL 7B LLM WHICH PROVIDED SIGNIFICANTLY BETTER ACCURACY FOR THE LABELS IT PROVIDED.

OUR SOLUTION

SCALABILITY

Used AWS Lambda functions to do all the core backend pipeline, which we know is highly scalable.

The LLM API calls are provided by AWS Bedrock which is a serverless, pay per demand type of service and is as scalable as Lambda functions with reliability promised by AWS.

The Dashboard uses an EC2 server which can be scaled using reactive scaling, or use a larger VM when the average demand increases.

Alternatively, the Dashboard server can be hosted on a private, company-owned, server.

KEY FEATURES

Smart Component Detection

Utilizes LLMs to intelligently classify issue content into seven distinct technical domains, spanning the entire stack from Frontend and Backend to Security and DevOps.

Issue Classification

Automatically categorizes issues into Bug, Enhancement, or Question.

Priority Assignment

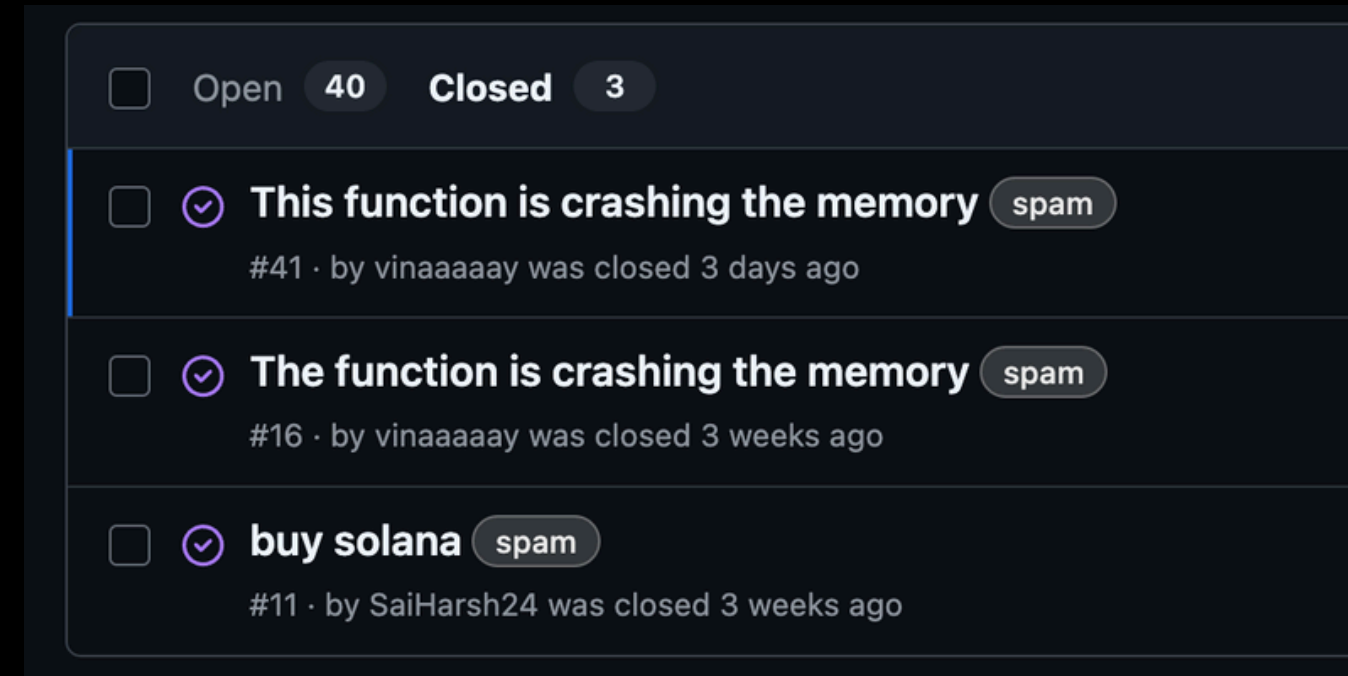
Analyzes urgency to assign priority levels (High, Medium, Low).

☐	Open	40	Closed	3	Author	Labels	Proj
☐		multivariate anomaly detection	backend	bug	High	SLA Breached	
		#43 · vinaaaaay	opened 3 days ago				
☐		Center Align Signin Form on Web Portal	enhancement	frontend	low		
		#42 · vinaaaaay	opened 3 days ago				
☐		ui not working	bug	frontend	High	SLA Breached	
		#40 · SaiHarsh24	opened last week				
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		#39 · SaiHarsh24	opened last week				
☐		driver trip direction is not updating correctly at start	backend	bug	High	SLA Breached	
		#38 · SaiHarsh24	opened last week				
☐		Driver Trip Direction Not Updating Correctly on Start	backend	bug	High	SLA Breached	
		#37 · vinaaaaay	opened last week				

KEY FEATURES

Spam Filtering

Automatically flags and closes spam or low-quality issues.

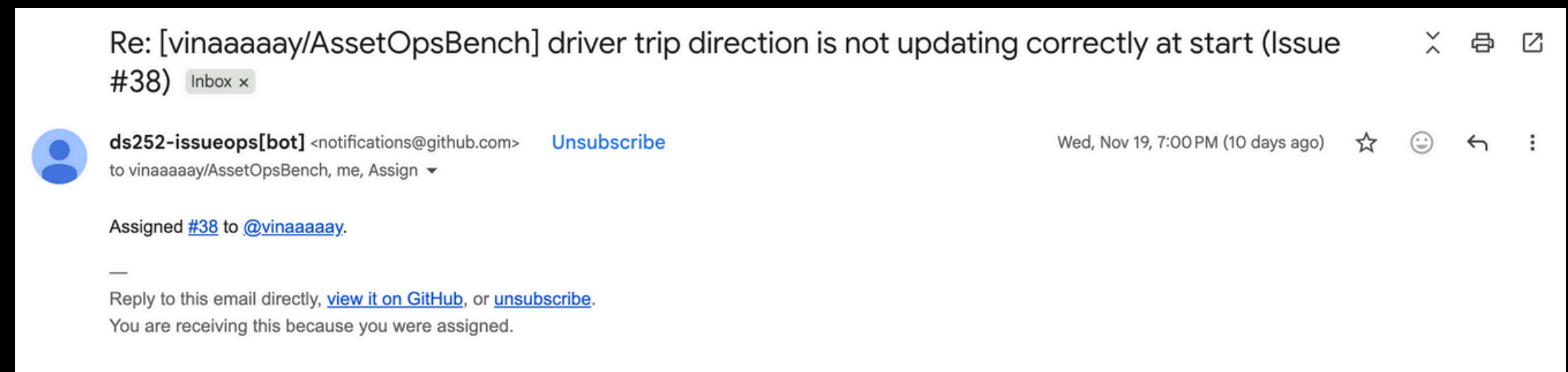


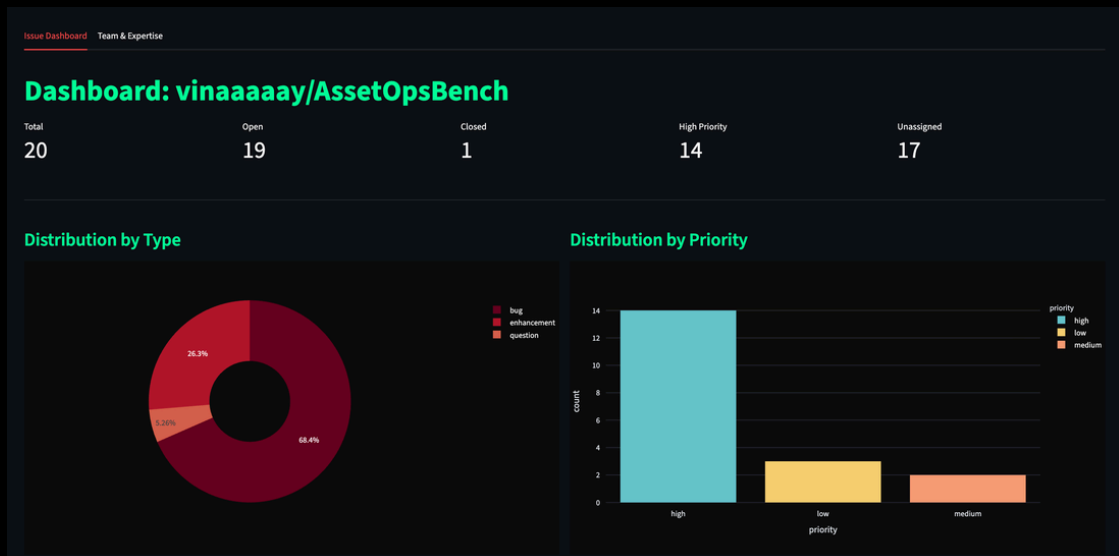
Expertise Matching

Routes issues to the best-suited maintainer based on their past contributions and skills.

Adaptive Workload Orchestration

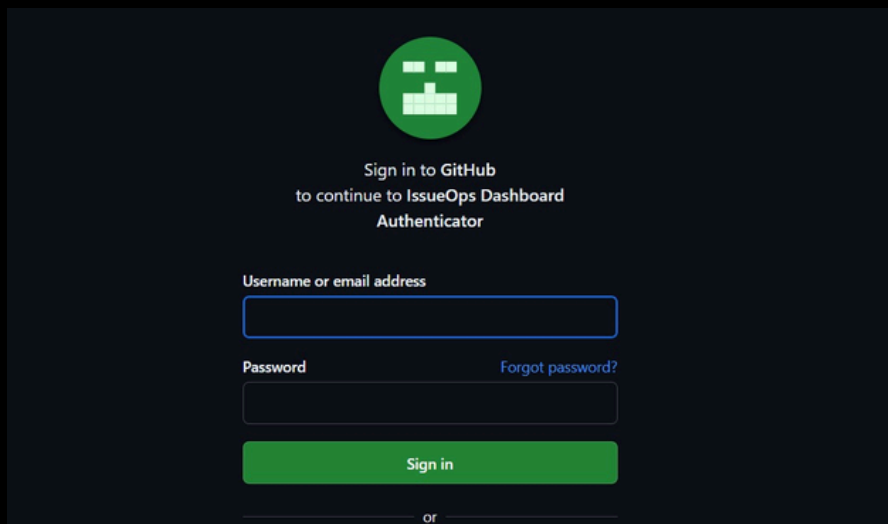
Synergizes continuous learning from recent contributor activity with real-time availability checks, dynamically routing tasks to the best-fit developers while enforcing working hours





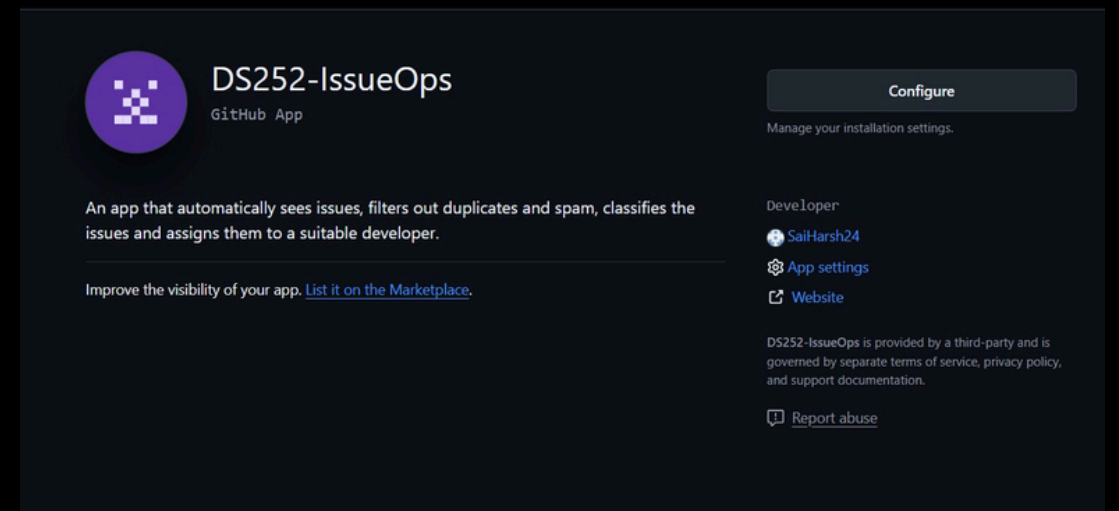
STREAMLIT DASHBOARD

Renders the interactive control plane for real-time monitoring, configuration, and manual pipeline triggers.



GITHUB OAUTH

Secures the admin dashboard by authenticating users and authorizing access based on existing repository permissions.



GITHUB APP

Serves as the secure, event-driven identity layer for integration and JWT-based authentication with repositories.



GITHUB API

Enables programmatic, bidirectional interaction to fetch raw issue data and execute final assignment actions directly on repositories.

UNDER THE HOOD

1. INSTALLATION PIPELINE

- Event-Based trigger to our system when a person installs the App.
- Automatically stores all the repos installed and triggers a scan to automatically infer Contributors's Expertise

2. ISSUE CLASSIFYING PIPELINE

- Gets issues from all the installed repos every hour, passes it through the pipeline and returns the classified labels to their respective repos.

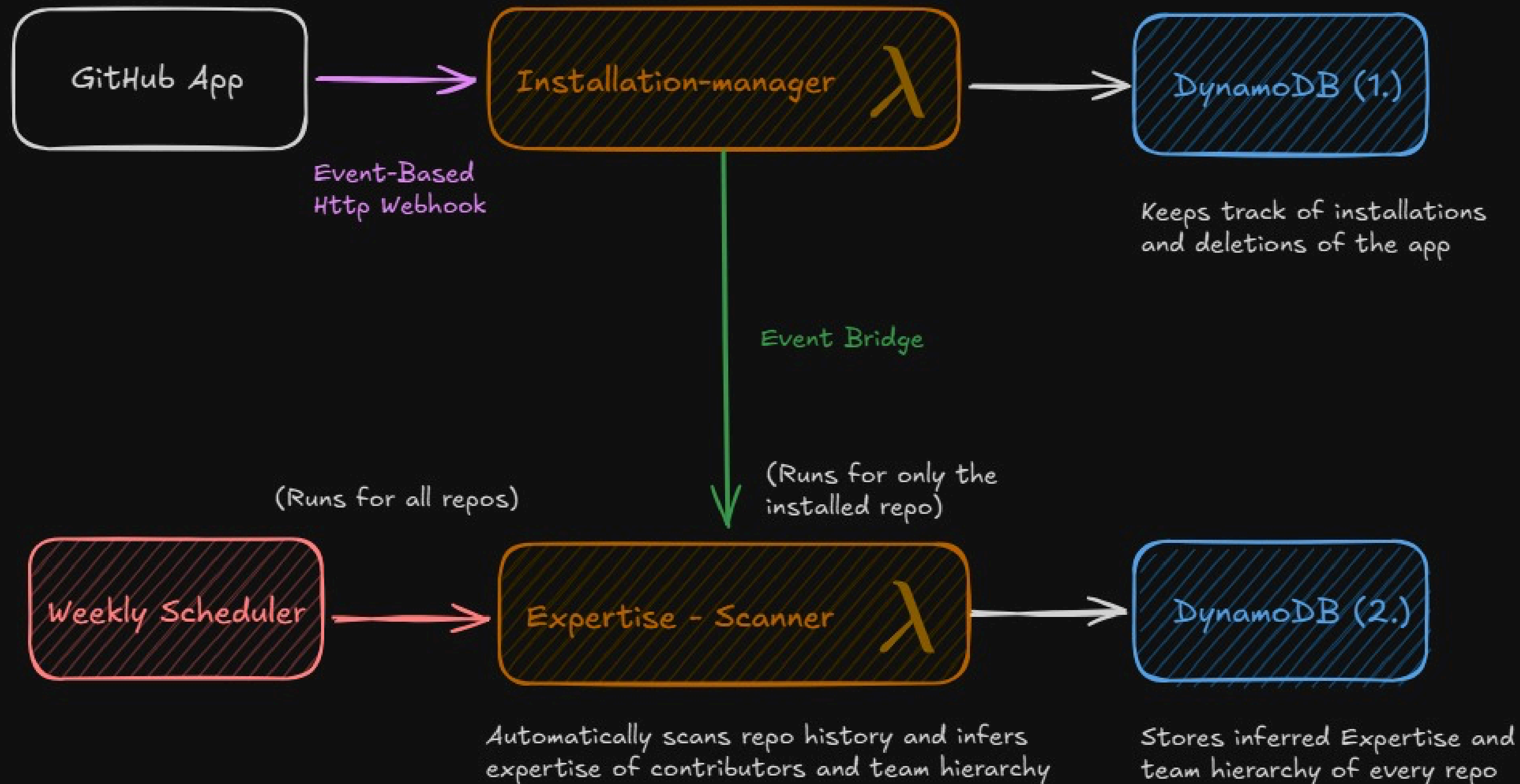
3. STALE AND SLA PIPELINE

- Scans the issues from installed repos and syncs it to the Dynamo Table to keep a track of which issues are closed and which do have no activity

4. DASHBOARD

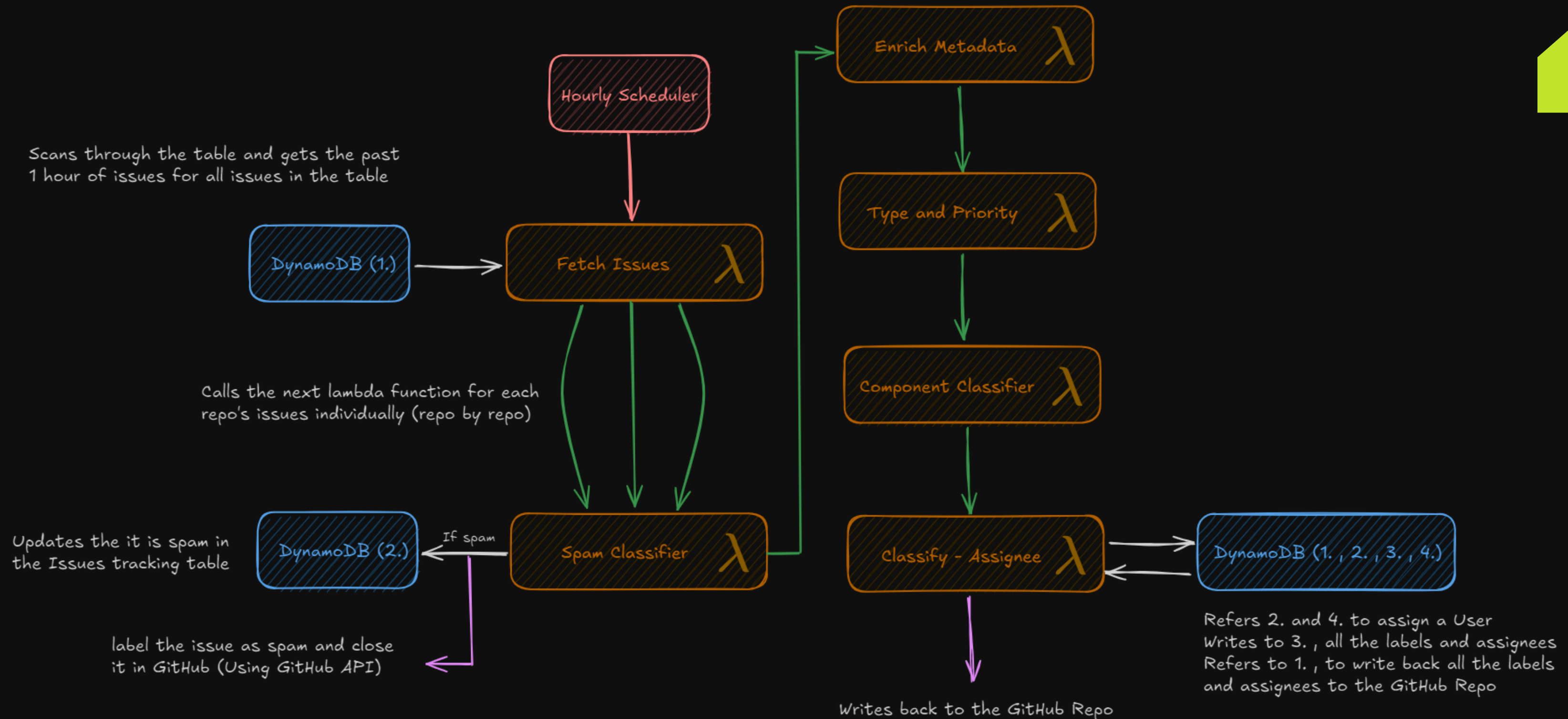
- A complete dashboard server that contains all the issues classified and your repos at one place.
- Allows to edit expertise and manual triggering of the classifying pipeline for more flexibility.

INSTALLATION PIPELINE



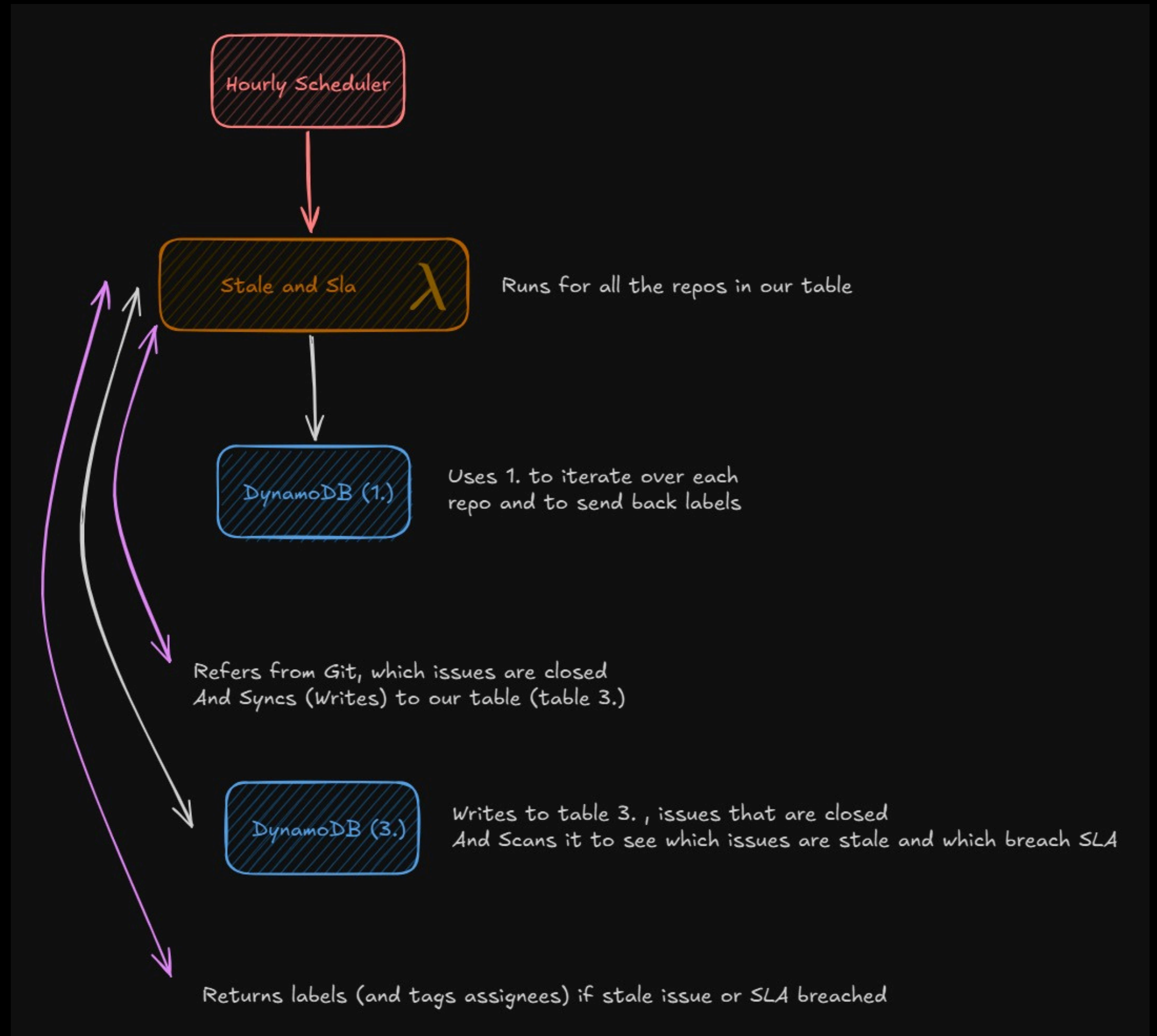
ISSUE CLASSIFYING PIPELINE

2



STALE ISSUE AND SLA PIPELINE

3



DASHBOARD

4



Shows you only your repos in which the app is included (using Table 1.)
Shows metrics of issues classified (using Table 3.)
Shows inferred expertise and timings which you can change



Gets issues only for the selected repo for the past custom number of hours, which can be chosen

From here the pipeline is the same

CORE CLOUD OVERVIEW

AWS IAM

Implements strict least-privilege security policies for every Lambda function and the dashboard instance.

AWS Lambda

Executes all core logic—from issue fetching to assignment and SLA enforcement—in a serverless, event-driven runtime.

EventBridge

Decouples the architecture by orchestrating asynchronous workflows and managing scheduled jobs for ingestion and expertise scanning.

Amazon DynamoDB

Acts as the single source of truth for tracking issue lifecycles, inferred developer expertise, and team availability.

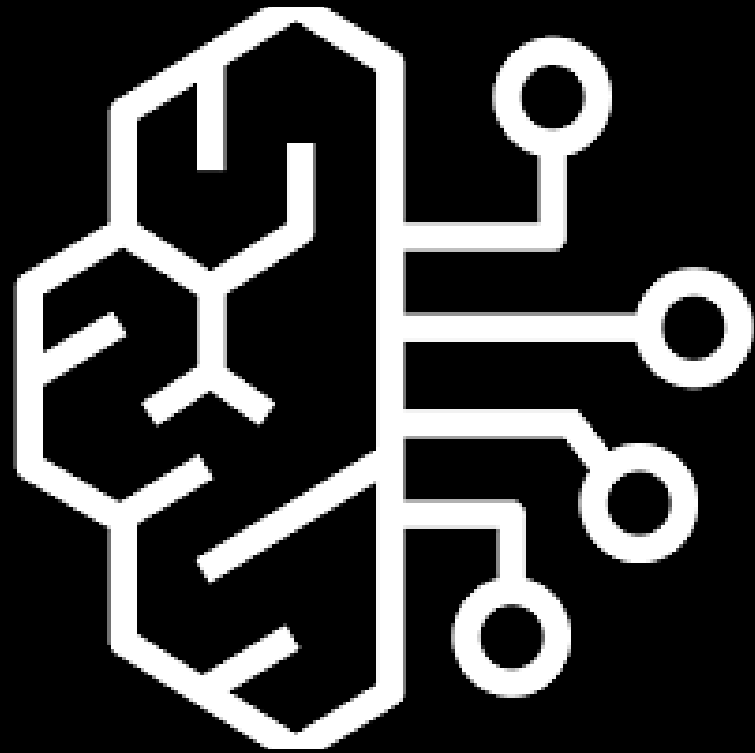
Amazon EC2

Hosts the Streamlit-based control plane and dashboard, providing a centralized interface for administration and metrics.

AWS Secrets Manager

Securely stores and manages the critical GitHub App private keys and OAuth client secrets required for authentication.

LLMs Used in Pipeline



AWS Bedrock

Serves as the scalable infrastructure backbone, providing unified access to the system's foundational models



Mistral 7B

Deciphers technical context to infer developer expertise and classify issue by type, priority and component.



Llama 3 8B

Optimizes data quality by filtering spam and refining issue metadata for semantic clarity.



DeepSeek V3

Powers the Master Agent reasoning core to synthesize insights and execute final assignment

AI Agents for Code Development



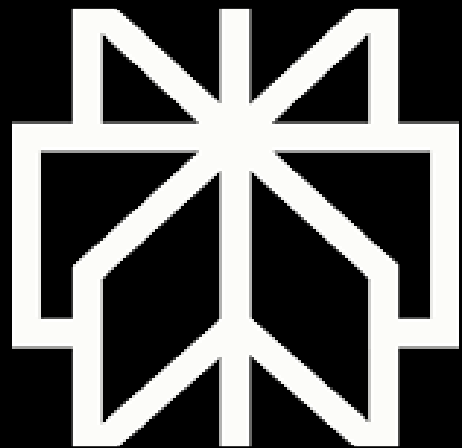
CHATGPT

Deeply structured reasoning, excels in complex queries.



GEMINI 3

Fast, context-aware insights, great for quick tasks.



PERPLEXITY

Accurate retrieval, but limited in synthesis capabilities.



CURSOR IDE

Strong code-aware refactoring, needs external reasoning aid.

38,023

lines of code

7,238

lines of comments

+

85%

code written by AI

120+

files created and integrated across backend, frontend, and pipelines.

Entire pipeline was manually deployed

HOW WE USED AGENTS

1. DISCUSSION

- We explored alternative approaches to achieve our project goals and held iterative discussions with AI agents to evaluate trade-offs and refine decisions.

2. CODE TUNING

- We provided the agents with rough offline testing code and asked them to optimise it, restructure it, and make it fully compatible with the overall pipeline.

3. DEPLOYMENT SUPPORT AND PIPELINE ALIGNMENT

- We supplied contextual documents and requested step-by-step deployment guidance.
- The agents generated deployment workflows for each Lambda function, configured EventBridge triggers, and updated the codebase to integrate DynamoDB updates and GitHub write-backs.

ORIGINAL TARGETS AND ACCOMPLISHMENTS

DELIVERED FEATURES (12/14)

- **Intake Agent:** Automated issue fetching via scheduled Lambdas & webhooks.
- **Spam Detection:** Llama 3 8B filtering at ingestion.
- **Stale Issue Handling:** Hourly "Housekeeper" for auto-closing/labeling.
- **Metadata Enrichment:** LLM-based title/body rewriting for clarity.
- **Classification:** Mistral 7B for Type (Bug/Feature) & Priority (High/Medium/Low).
- **Component Detection (Bonus):** Identifies affected areas (Frontend/Backend).
- **Skill Matching:** expertise_scanner builds dev skill graphs from commit history.
- **Availability Check:** infer-availability calculates UTC working hours from activity.
- **Workload Balancing:** Real-time open issue checks to prevent overload.
- **SLA Escalation:** Auto-pings assignees on high-priority breaches (>54h).

MODIFIED SCOPE

- **Duplicate Detection:** Deprioritized due to complexity (vector DB requirements).
- **Notification Agent:** Replaced custom bot with native GitHub @mentions for better integration.
- **TO DO Tracking Agent:** Skipped as it required reading full repository source code, which was out of scope for this metadata-focused project.

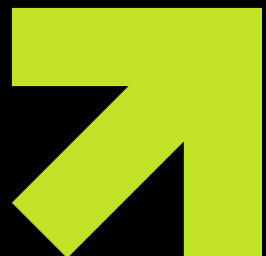
PEER TESTING RESULTS

WE RECEIVED FOUR ISSUES DURING PEER TESTING, PRIMARILY FOCUSED ON INCORRECT ISSUE CLASSIFICATION.

THE ROOT CAUSE WAS UNRELIABLE PERFORMANCE FROM OUR LOCAL CLASSIFICATION MODELS.

WE REPLACED THEM WITH AWS BEDROCK INFERENCE AND ADOPTED THE MISTRAL-7B MODEL FOR EFFICIENT, ACCURATE CLASSIFICATION

ANOTHER ISSUE HIGHLIGHTED CONFUSION IN THE README AND ONE OF THE FEATURES, WHICH WE RESOLVED BY REFINING DOCUMENTATION AND CLARIFYING THE WORKFLOW.



THANKS TO THE TESTERS BHARGAV SASHANK AND HARSH SAXENA FOR OPENING ISSUES AND TAKING THE TIME TO TEST THE GITHUB APP

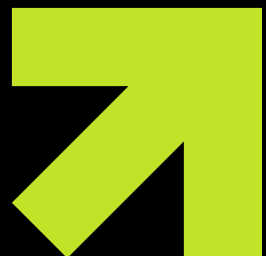
FUTURE WORK

DEVELOP IN-HOUSE LIGHTWEIGHT MODELS FOR ALL CLASSIFICATION TASKS TO REDUCE EXTERNAL DEPENDENCIES AND OPTIMIZE COST.

IMPLEMENT ROBUST DUPLICATE-ISSUE DETECTION TO FURTHER STREAMLINE TRIAGE AND REDUCE NOISE.

INTEGRATE THE SYSTEM WITH JIRA TO SUPPORT ENTERPRISE WORKFLOWS BEYOND GITHUB.

ADD CONFIGURABLE AND MANUALLY ADJUSTABLE SLA RULES FOR FLEXIBLE PRIORITIZATION AND ESCALATION POLICIES.



THANK YOU